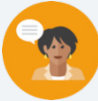


12.6 Lengths of Secants, Tangents, Segments, and Chords in Circles

Lesson Introduction

 Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems involving the length of segments of a circle formed by intersections.
 Benchmark Coherence	Building On: N/A		Working Toward: N/A	
 Essential Questions	What is the difference between a chord, secant, and tangent line to a circle?			
 Lesson Narrative	This lesson will bring together all the properties of chord, secant, and tangent lines in a circle. During this lesson, students will be matching each description with the correct image, type of line segment, and answer. Students will need to be able to differentiate between the types of line segments of circles in order to use the correct method of solving.			
 Component Summary	12.6.1 Warm-Up (~5 min.)	Consider how math is a mode of communication (SE p. 257)	12.6.3 Try It (10-15 min.)	Real-world problems solving for length (SE p. 259)
	12.6.2 Activity (30-35 min.)	Matching activity with line segments of circles (SE p. 258)	12.6.4 Wrap-Up (~5 min.)	Students summarize their learning (SE p. 260)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Chord - Secant - Tangent		- Match Sheet Activity Pages (Optional) Poster or Chart Paper	If you want to pre-cut the math sheet activity pages, students could group them together on their desk for each question.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse the difference between a chord, secant, and tangent line in a circle. - Students may forget the different formulas to solve to find a chord, secant, or tangent line in a circle. - Students may miscalculate when using the formulas, leading them to an incorrect answer.	Consider having all the different line segments of a circle with each corresponding formula and method of solving posted throughout your room for students to use as a reference.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 12.6.2 Activity is a card matching activity. Students have quiet time to think about the activity and work on it individually. Then, they take time to share their responses or their progress with a partner. If time allows, once these partner conversations have taken place, some partnerships can be selected to share their thoughts with the class.	MA.K12.MTR.3.1: Mathematical Fluency This lesson is an opportunity for students to complete several tasks with mathematical fluency. They will need to select efficient and appropriate methods for solving these questions, given the different lines of the circle. The goal of the lesson is for them to be able to complete the different question types with accuracy and confidence.
 Differentiate Instruction	12.6.2 Activity is meant to be completed individually. If you anticipate that students might struggle during this activity, consider having the different types of line segments of circles posted in your room on posters and have the students do a Gallery Walk before starting this lesson. At each type of line segment of circles, have students write down on the poster things they know about the line segments, examples, and/or drawings. Provide a visual entry-point from seeing the connection between the description, type of line segment, and the image. Provide an auditory entry-point from the collaborative conversation at each poster. Provide a kinesthetic entry-point from moving around the room from poster to poster.	
Lesson Notes:		

12.6.1 Warm-Up: Critical Thinking

Component Overview: Students are asked to use critical thinking to answer questions.



12.6.1 Warm-Up: Critical Thinking

1. Explain how math could be considered a language used for communication.

Answers may vary

*Math, like other languages, is a method of communicating.
It consists of structure, symbols, and rules.*

2. Suppose you were taking Geometry in another country.
How do you think the problems would differ?

Answers may vary.

The vocabulary would be different.



Do you have students who speak other languages in your classroom? Consider inviting a guest speaker who speaks a foreign language to join class (in person or virtually). Additionally, consider having students return to this lesson later in the year for project-based learning around this topic.

12.6.2 Activity: Line Segments of Circles

Component Overview: Students tie together the different line segments of circles studied in Unit 6, including: chords, secant lines, and tangent lines.



12.6.2 Activity: Line Segments of Circles

Match each image, type of segment, and answer to the description of circles. Record your answers in the table.

Circle	Work	Cards' Letter
Circle <i>K</i> has points <i>G</i> , <i>N</i> , <i>D</i> , and <i>E</i> on the circle, where $GM = (2x + 2)$, $MD = 14$, $EM = 12$, and $MN = (2x + 5)$	$(2x + 2)(14) = (2x + 5)(12)$ $x = 8$	Image: <u>W</u> Segment: <u>E or D</u> Answer: <u>M</u>
Circle <i>H</i> has points <i>C</i> , <i>T</i> , and <i>B</i> on the circle, where $BR = 15$, $RT = 9$, and $CT = x$	$15^2 = 9(x + 9)$ $x = 16$	Image: <u>N</u> Segment: <u>F</u> Answer: <u>K</u>
Circle <i>R</i> has points <i>T</i> , <i>S</i> , <i>V</i> , and <i>W</i> on the circle, where $TS = (x - 8)$, $SP = 4$, $VP = 3$, and $VW = (x - 5)$	$(x - 8 + 4)(4)$ $= (x - 5 + 3)(3)$ $x = 10$	Image: <u>P</u> Segment: <u>B</u> Answer: <u>H</u>
Circle <i>F</i> has points <i>E</i> and <i>B</i> on the circle, where $EY = (7x - 3)$ and $BY = (5x + 1)$	$7x - 3 = 5x + 1$ $x = 2$	Image: <u>R</u> Segment: <u>A</u> Answer: <u>G</u>
Circle <i>V</i> has point <i>B</i> on the circle, where $VB = 4$, $BA = 8$ and $VA = x$	$4^2 + 8^2 = x^2$ $x = 4\sqrt{5}$	Image: <u>S</u> Segment: <u>C</u> Answer: <u>L</u>
Circle <i>A</i> has points <i>G</i> , <i>R</i> , <i>H</i> , and <i>S</i> on the circle, where $MA = WA$, $GR = 35$, and $SH = (6x - 7)$	$6x - 7 = 35$ $x = 7$	Image: <u>T</u> Segment: <u>E or D</u> Answer: <u>J</u>



Group Presentation

Consider assigning each group one of the problems in 12.6.2 Activity. Each group will present to the class by explaining which line segments are used, which image matches the circle, and justifying the answer. Students can use this time to ask clarifying questions, and ask members of each group to restate their understanding of the group's findings in their own words.



Consider challenging students by removing the answer cards. Students can verify their answers at the end with a partner.

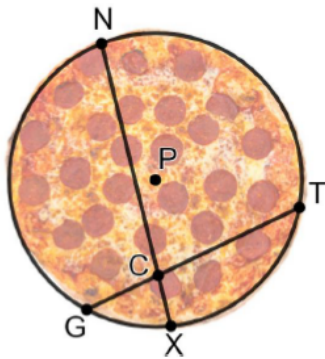
12.6.3 Try It: Solving for Length

Component Overview: This component positions students to make connections between chords, secants, and tangent lines in mathematical and real-world contexts.



12.6.3 Try It: Solving for Length

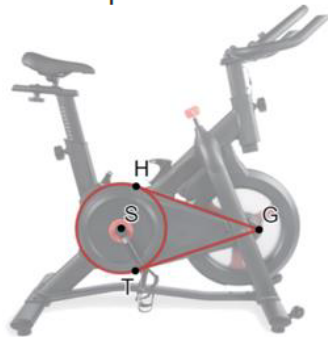
1. When Doug's pizza was delivered, he realized the pizza was not cut evenly. The pizza, circle P , is cut by chords $\overline{NX}$ and $\overline{GT}$.



Given that $NC = 5x - 6$ inches (in), $CX = 2$ in, $GC = 3$ in, and $CT = 2x$ in, what is the length of $\overline{NX}$?

$NX = 11$ in.

2. A belt guard is used on stationary bikes to cover some of the inner mechanics. Circle S is shown, where points H and T are on the circle. Tangents $\overline{HG}$ and $\overline{TG}$ intersect outside the circle at point G .



If $HG = 15$ and $TG = 4y - 9$, what is the value of y ?

$y = 6$



Use questioning techniques and methods to create a student-centered learning environment in which students are driving the learning. Consider using sentence stems to encourage conversations:

- "I had to pay attention to how I labeled the diagram because..."
- "I know the line segments in this question are... because..."
- "I needed to make sure I was using the correct formula because..."

12.6.4 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



12.6.4 Wrap-Up: Cool Down

1. Write **one question** you still have about solving mathematical and real-world problems involving the lengths of secants, tangents, segments, or chords of a circle.

Answers may vary.



Lessons 7-9 explore the equation of a circle. Consider utilizing EdgeXL for a summative assessment of Lessons 1-6 to determine student progress regarding secants, tangents, and problem-solving with circles.